Natural Sciences: Photosynthesis Learning Pack

classroom_bundle | Grade 7 learners and educators

Product	homs
Subject	Natural Sciences
Grade	7
Curriculum focus	Term 1: Photosynthesis and plant nutrition
Language	English
Version	1.0.0

CONTROLLED REVIEW ASSET. Human approval is required before external release.

Contents

Update this field in Word or LibreOffice to populate the table of contents.

Learning objective: Explain how green plants use light energy, carbon dioxide and water to produce glucose and oxygen.

1. Core concept

Photosynthesis is the process by which chlorophyll-containing plants convert light energy into stored chemical energy. The process takes place mainly in the leaves.

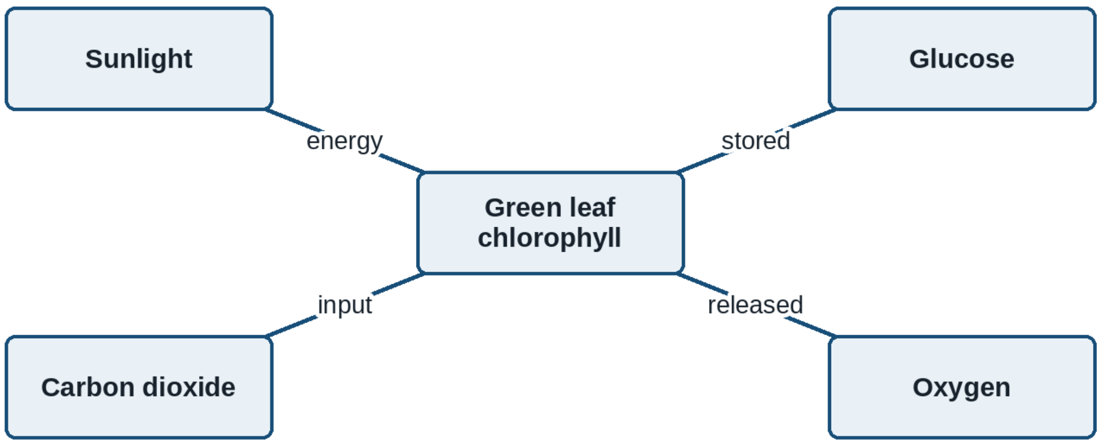

Figure 1: Inputs and products of photosynthesis

```
carbon dioxide + water -> glucose + oxygen  (in the presence of light and chlorophyll)
```

2. Guided investigation

Teacher note: Prepare one destarched plant and demonstrate the iodine test using the school's approved laboratory safety procedure.

Learner instruction: Record the colour observed before and after iodine is added. Do not handle hot ethanol or an open flame.

Investigation observations

Leaf area	Light exposure	Colour after iodine	Starch present?
Covered section	No direct light	Yellow-brown	No
Exposed section	Direct light	Blue-black	Yes

2.1. Identify the independent variable in this investigation. **(1)**

2.2. Use the observations to explain why light is necessary for starch production in the leaf. **(3)**

3. Assessment rubric

Explanation rubric

Criterion	2 marks	1 mark	0 marks
Evidence	Uses both observed colours accurately	Uses one relevant observation	No relevant observation
Reasoning	Links light to starch production	States a partial link	No valid scientific link

Annexure A: Educator memorandum

Expected response: 2.1 Light exposure. 2.2 Only the exposed area turned blue-black, showing that starch formed where light was available.

Plants do not absorb food from soil. They manufacture glucose through photosynthesis.